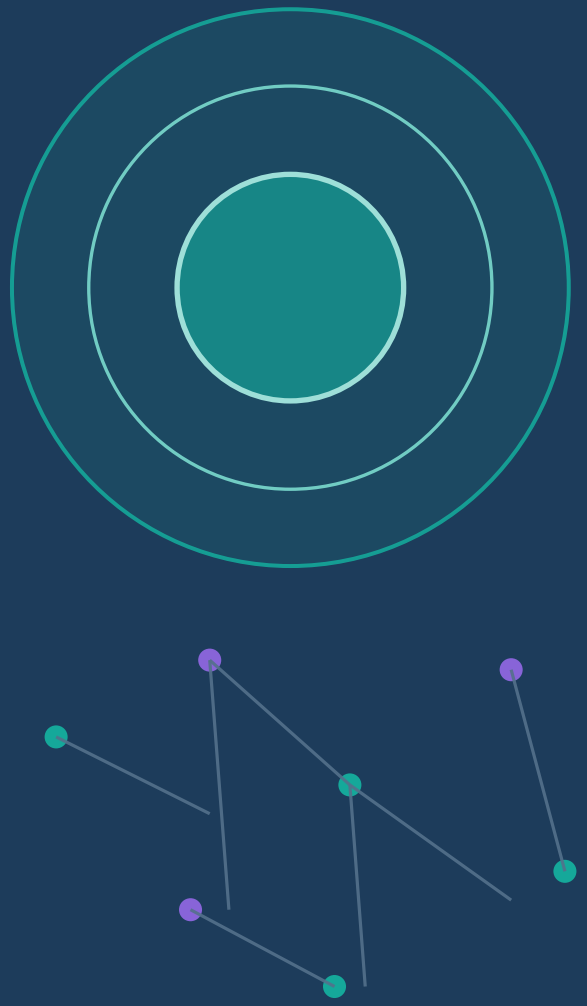
HAPPIEST HEALTH / DIGITAL TWIN CHALLENGE 2026

AnxietyTwin

Personalized anxiety monitoring through an adaptive digital twin

A prototype connecting longitudinal observations, scenario simulation, explainable ML research, and grounded AI support.

PROJECT PRESENTATION



Research prototype · Not a diagnostic or clinically validated system

From fragmented signals to a personal, explainable picture

AnxietyTwin is designed to help users explore patterns and possible responses—not to diagnose or replace professional care.

OBSERVE

Check-ins, self-reported distress, sleep/activity signals, and longitudinal history.

MODEL

A digital twin represents a personal baseline and supports controlled what-if exploration.

INTERPRET

Dynamic-feature ML research estimates next-day DBP risk; rules add contextual indicators.

SUPPORT

Grounded AI assistance, coping exercises, privacy controls, and optional clinician review.

CORE VALUE

A single experience for tracking, simulation, explainable predictions, and reflective next steps.

Evidence posture: promising experimental signal under participant-aware evaluation; clinical utility and real-world benefit remain unvalidated.

Anxiety is personal, dynamic, and difficult to summarize

A daily record can be useful, but it rarely tells the whole story on its own.

FRAGMENTED SIGNALS

Distress, sleep, activity, and physiological measures may live in separate views—or not be recorded consistently.

CHANGING BASELINES

The same signal may mean different things for different people and in different contexts.

LIMITED EXPLORATION

Users need a safe way to reflect on hypothetical scenarios and coping choices without treating a model output as certainty.

DESIGN GOAL

Connect observations + personalized simulation + cautious decision support.

Scope is non-diagnostic wellness and research support—not automated clinical decision-making.

A connected user journey—not just a prediction score

The experience brings data, simulation, interpretation, and reflection into one workflow.

01

CAPTURE

Check-ins, reported context, available observations

02

PERSONALIZE

Personal baseline + twin memory over time

03

EXPLORE

What-if parameters, scenario replay, synthetic streams

04

UNDERSTAND

Explainable trends, rules, next-day DBP risk estimate

05

RESPOND

Coping exercises, notes, goals, optional clinician review

Cross-cutting principles **Consent • Data provenance • Clear uncertainty • Safety guardrails • No diagnostic claims**

What the prototype brings together

Core experiences reported as implemented; availability and final behavior should be verified in the live build.

Tracking & reflection

Dashboard, check-ins, timeline/history, goals, journal, weekly report.

Adaptive digital twin

Personal baseline, twin memory, what-if inputs, comparison timelines, deterministic replay.

Scenario lab

Start/pause/resume/stop, speed controls, bounded timeline, synthetic HR/HRV/sleep/distress streams.

Insights & interventions

Rule-based indicators, explainable patterns, contextual follow-up prompts, coping exercises.

Clinician workspace

Consent-based access grants, observation timeline, review notes and follow-up tasks.

Trust & control

Privacy and consent controls, data-quality view, data provenance and PDF export.

A layered system with explicit AI boundaries

Conceptual overview based on project records; validate exact DB/deployment settings against the current repository.



USER / DOMAIN DATA

Observations, consent, twin snapshots, clinician review records. Synthetic scenarios remain distinguishable from user-reported history.

ML MODEL ARTIFACT

Production Random Forest pipeline (`production_model.joblib`) with 35 dynamic input features.

AI KNOWLEDGE PATH

Assistant tools are authorized server-side; RAG retrieves from clinical documents; an LLM provider is optional.

Data flow **Web UI → authenticated FastAPI → domain services → database / ML / AI services → explainable UI response**

The LLM layer does not replace the specialized prediction model. Without a configured provider, assistant responses use a deterministic grounded fallback.

A research dataset with a deliberately narrow prediction target

Dataset statistics and target semantics come from project audit reports.

3,969	43	254	~6.4%
participant-days	participants	positive DBP labels	label prevalence

PREDICTION TARGET

Day Before Panic (DBP): estimated risk for the following day, approximately a 24-hour horizon.

WHAT IT DOES NOT PREDICT

The data does not support reliable prediction of a panic episode in the next few minutes or hours; the model is not a clinical diagnosis.

IMBALANCED LABELS

With only about 6.4% positive days, PR-AUC and calibration matter alongside ROC-AUC; a high ROC-AUC alone is not proof of practical utility.

The key ML outcome was discovering and addressing leakage

Random day-level splitting made the task look substantially easier than generalizing to unseen participants.

INITIAL WARNING

Random split ROC-AUC was about 0.90, but participant-aware validation fell sharply. Static participant/clinical fingerprints could leak identity-related signal.

AUDIT & REPAIR

The pipeline was reworked around dynamic features, fold-contained preprocessing, participant GroupKFold, and chronological holdout evaluation.

VALIDATION POSTURE

Report cross-participant and future-time metrics separately. Full-dataset resubstitution scores are in-sample and must not be presented as independent validation.

Lesson learned

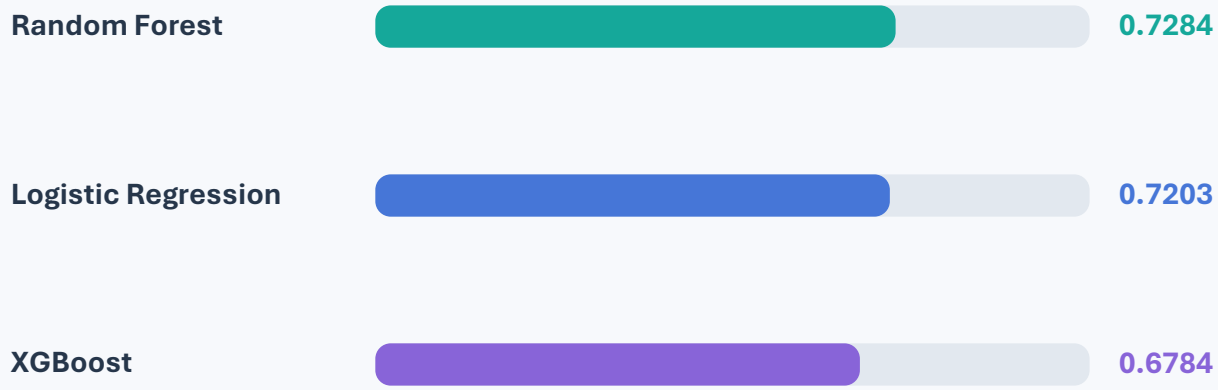
Validation design is part of the model—not a final cosmetic check.

The production model uses 35 dynamic features; 40-feature personalization experiments are research-only and are not the deployed endpoint.

Participant-aware evaluation gives a more honest performance picture

Primary reported GroupKFold results for dynamic features; these are not clinical validation results.

5-fold GroupKFold · unseen-participant generalization



RF ROC-AUC

0.7284 ± 0.1024

RF PR-AUC

0.1827 ± 0.052

Brier score (benchmark report)

0.0932

Scores vary across project reports/evaluation setups; use the benchmark report values above for GroupKFold comparisons. Chronological personalized experiments are separate results.

A chronological test separates future days from training history

Reported Experiment C uses a time-based split rather than scoring the same rows used to fit the model.

EARLIER PERIOD · TRAIN

2,251 days · 178 positives

FUTURE PERIOD · TEST

1,517 days · 73 positives

POPULATION RF

ROC-AUC 0.8272 · PR-AUC 0.3463 · Brier 0.0918

PERSONALIZED RF (RESEARCH)

ROC-AUC 0.8504 · PR-AUC 0.4574 · Brier 0.0477

Important calibration caveat

On this holdout, the naive prevalence baseline Brier score was 0.0458—better than personalized RF. Personalized XGBoost Brier 0.0396 was better in this experiment, but remains research-only.

Keep the shipped model distinct from research experiments

The production endpoint and the personalized research evaluation are not the same model configuration.

PRODUCTION PIPELINE

Random Forest · 35 dynamic features
Artifact: `ml/models/production_model.joblib`
Used by backend ML service; intended for next-day DBP risk.

RESEARCH EXPERIMENTS

40-feature personalized models use longitudinal context and have separate evaluation results. They are not the deployed production endpoint.

DO NOT CLAIM

The in-sample ROC-AUC 0.964 as real-world accuracy, or the experimental personalized score as production performance.

A next step is external prospective validation, pre-registered thresholds, calibration monitoring, and evaluation of clinical usefulness before deployment in care.

Scenario simulation makes the twin interactive and inspectable

The live scenario engine supports exploration of trajectories and interventions in a controlled prototype.

INPUTS

Baseline context, hypothetical changes, distress feedback, synthetic HR/HRV, sleep and activity patterns.

SIMULATION

Deterministic scenario generation, playback controls, time-bounded timeline, baseline comparison and replay.

REFLECTION

Compare scenarios and pre/post distress, try coping exercises, and save longitudinal snapshots where supported.

Transparency rule

Synthetic simulator outputs are demonstrations—not sensor measurements and not validated predictions of what a person will experience.

WHAT-IF EXPLORATION

REPLAY & COMPARISON

BASELINE EVOLUTION

USER-CONTEXT CHECKS

A large language model complements—not replaces—the predictive model

The assistant architecture combines provider abstraction, scoped tools, retrieved knowledge, and safe fallback behavior.

USER QUESTION

Conversation in the assistant drawer; identity comes from authenticated session.

AUTHORIZED TOOLS

Baseline, recent observations, trends, prediction explanation, coping, knowledge query.

RAG RETRIEVAL

Searches a local clinical knowledge collection; retrieved text grounds relevant answers.

RESPONSE LAYER

Provider adapter + crisis guardrails + deterministic grounded fallback if no LLM configured.

Reported provider options

Gemini | OpenAI | Anthropic | Local Ollama

LLM truth-in-labeling — **Without a configured API/local provider, the fallback is deterministic—not a general-purpose LLM.**

Trust controls are core system components

Because the domain is sensitive, the prototype must keep the user in control and expose data provenance clearly.

Consent-first sharing

Clinician view is intended to require explicit access grants.

Identity-scoped tools

Assistant tool access derives identity from authenticated context, not user-supplied prompt IDs.

Data provenance

Mark synthetic scenario data separately from historical and user-reported observations.

Crisis-aware UX

Guardrails should point to suitable support resources; local contacts must be verified for the user's region.

Transparent language

Show uncertainty, model horizon, missing data and limitations; avoid diagnostic claims.

Audit & retention

Review deletion behavior, audit logs, exports and access isolation before public release.

Before production healthcare use: independent security, privacy, safety and clinical validation are required.

The work moved from feature construction to verification and hardening

Implementation and test counts below are reported by project audit summaries, rather than independently re-run here.

BACKEND

FastAPI services, database models, prediction endpoint, scenario engine, assistant tools, consent/access paths.

FRONTEND

Dashboard, twin/simulation UI, clinician route, assistant drawer, reports, privacy and data-quality pages.

AUTOMATED CHECKS

Latest report in project notes claims 82/82 backend tests and 79/79 frontend tests at the AI-upgrade stage; a later release report claims 73 backend and 74 frontend tests. Counts differ by run/report.

Verification discipline

Re-run tests and production build from the final commit before recording or submitting.

Treat reported implementation status as project notes until confirmed against the final repository and deployed demo.

What the current prototype can—and cannot—claim

Clear boundaries make the work more credible and protect users from overinterpreting model outputs.

WHAT IT CAN SHOW

Personal tracking, longitudinal patterns, hypothetical scenarios, explainable experimental DBP risk estimates and grounded wellness support.

WHAT IS NOT ESTABLISHED

Clinical effectiveness, diagnostic accuracy, causal effects, imminent panic prediction, broad population generalization or improvement in patient outcomes.

- Small cohort: 43 participants and low positive-label prevalence (~6.4%).
- Generalization varies across participants; confidence/calibration and prevalence baselines matter.
- Rule thresholds are heuristic and not clinically validated.
- Simulated biometrics are synthetic examples—not live wearable data.

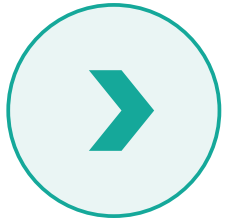
The path forward is validation-first, not feature-first

Prioritize evidence, repeatability, and user safety over adding more interface surface.

01	**REPRODUCIBILITY**	Pin environment, data version, model artifact and seeds; rerun tests from a clean checkout.
02	**EXTERNAL VALIDATION**	Use prospective and independent cohorts; predefine splits, targets and evaluation protocol.
03	**CALIBRATION & UTILITY**	Compare to prevalence baselines; select thresholds on training data; monitor calibration and false alerts.
04	**PRIVACY & SAFETY**	Audit consent/revocation, tenant isolation, deletion, export, logs and regional crisis resources.
05	**USER STUDY**	Assess comprehension, usability, trust, burden and whether insights help users reflect safely.

A guided demo should tell one coherent story

Demo video link supplied for the project: use it as the walkthrough anchor and verify its visibility before submission.



DEMO LINK https://youtu.be/NQzYV4Sl_rw

Suggested walkthrough order

- Dashboard & check-in
- Twin what-if / scenario replay
- Interpretation & AI assistant
- Privacy / clinician workspace
- Close with ML scope & limitations

I have not independently viewed the video playback. Confirm it is accessible and that the required duration/visibility match the challenge rules.

AnxietyTwin

A digital twin for reflection,
not a diagnosis.

The contribution is the integrated experience—and the disciplined work to make its predictions more honest, explainable and safe.

Thank you

Research prototype · clinical validity not established